Hindsight Agent: A Multi-Agent System for Quantitative Trading

Rohit Edward Martires
Independent researcher, Goa, India
rohitmartires14@gmail.com

Abstract

Large language models are increasingly used in financial decision pipelines, yet published agent stacks are often hard to reproduce, rarely enforce temporal data boundaries, and seldom report net returns after transaction costs. We present **Hindsight**, an open LangGraph stack with structured Pydantic outputs, a simulation end date for point-in-time data, ticker anonymization to avoid any LLM bias for company names, and a fee-aware paper backtest suite.

The primary contributions are reproducible systems engineering, auditable stage artifacts, and frozen CSV schedules. We then run the full pipeline on RELIANCE.NS and TCS.NS, one pre-specified bear window and one bull window each, from $100,000. In both bear windows the agent loses less than buy-and-hold and draws down less. A 10/50 moving-average rule, with the same fees and no language model, does the same and loses even less. In both bull windows the agent trails buy-and-hold. The single RELIANCE run trails by 9.5 percentage points. On TCS the first run trails by 2.3 percentage points; five further runs of that window trail by 14.1 on average. The other three windows are one run each. **Code and artifacts:** https://github.com/RMartires/hindsight.

1. Introduction

Researchers are experimenting with large language models (LLMs) inside trading workflows amid a rapid expansion of LLM-driven agent systems. Yet evaluations rarely ask how the same implementation behaves when the underlying trend differs, rarely enforce point-in-time data access at the adapter layer, and rarely report net returns after stated transaction costs.

What this paper is. Hindsight extends the open-source TradingAgents multi-agent LangGraph framework (Xiao et al., 2024) with (i) schema-constrained stage outputs, (ii) a simulation end date for point-in-time vendor queries, (iii) ticker anonymization, and (iv) a scriptable, fee-aware paper backtest suite tied to frozen CSV artifacts. We document how to build and audit such a stack and provide a scoped, pre-specified regime-split case study on two names.

What this paper is not. We do not claim state-of-the-art risk-adjusted returns, regime prediction, or statistical generalization from a broad equity universe. Seed variance is reported for TCS bull (five runs). The other three windows are one trajectory each. Frontier empirical papers (e.g., Jeon and Lee, 2026; Miyazaki et al., 2026) average tens of random seeds across index universes; our evaluation scope is deliberately narrower and is labeled as such (§4.1, Table 1).

Case-study evaluation. Each window starts from $100,000. In both bear windows the agent loses less than buy-and-hold and draws down less. A 10/50 moving-average rule does the same, and loses even less, so that pattern is not specific to the agent (Table 3). Both bull windows trail buy-and-hold. RELIANCE trails by 9.5 pp on its only run. On TCS the first run trails by 2.3 pp; five further runs average −14.1 pp and also finish behind the rule (§4.4).

2. Related Work

LLM agents and classic MAS. Multi-agent frameworks orchestrate specialist roles for trading and finance (Xiao et al., 2024; Yu et al., 2023; Yang et al., 2023). Bădică et al. (2025) situate this wave as a shift from explicitly programmed BDI-style agents to emergent LLM behaviors, raising new demands for verifiable coordination and audit trails. While **TradingAgents** (Xiao et al., 2024) provides the foundational analyst → debate → trader graph that we extend, it does not ship adapter-level temporal clamps, schema-persisted stage artifacts, or a fee-aware replication harness. **AlphaAgents** (Zhao et al., 2025) validates institution-style role separation (Fundamental, Sentiment, Valuation) with structured debate; **QuantAgents** (Li et al., 2025) emphasizes manager-led meetings and simulated trading feedback; both are comparators for our Bull/Bear debate loop, but neither foregrounds point-in-time vendor routing or frozen schedule CSVs as first-class outputs.

Methodological standards, leakage, and overfitting are central challenges in current benchmarks. Nguyen and Pham (2026) survey existing LLM financial multi-agent frameworks, exposing widespread vulnerabilities to transaction cost neglect and regime-shift blindness. To establish rigorous cross-system comparisons, they propose a set of five Consolidated Minimum Standards and introduce the **Coordination Breakeven Spread** (CBS) to explicitly measure the net-of-cost alpha generated by multi-agent coordination relative to trading drag. We report a simplified fee-and-turnover breakeven in §4.5 and defer full latency CBS to future work. Bailey et al. (2014) show that undisclosed configuration search makes strong in-sample backtests statistically cheap to manufacture; Hindsight responds with pre-specified windows marking different regimes, frozen artifacts, and seed variance on one window rather than implied generalization.

Point-in-time policy and anonymization. Early LLM trading papers often relied on prompt-level "do not peek" instructions. Nguyen and Pham (2026) treat contamination control and net-of-cost returns as minimum standards. Hindsight implements the former via a simulation end date at vendor routing and the

latter via a Zerodha delivery paper ledger. Jeon and Lee's **BlindTrade** (2026) provides the strongest anonymization-first defense: tickers like "AAPL" can trigger memorized pretraining associations rather than fresh reasoning; they anonymize S&P 500 constituents daily and report Sharpe 1.40 ± 0.22 across 20 seeds. Hindsight's `STOCK_####` aliases (§3.4) pursue the same goal within a LangGraph replication stack.

Fine-grained tasks and coordination topology. Miyazaki et al. (2026) show that fine-grained analyst tasks (e.g., explicit ratio and indicator computations) outperform coarse "analyze fundamentals" prompts under leakage-controlled TOPIX-style evaluation with 50 trials. Our market and fundamentals analysts are tool-bound to concrete fetches and Tier-1 indicators (§3.2), aligning architecture with that finding without claiming comparable universe-scale statistics. Dewansh (2026) frames explainable stability and rationale persistence as audit criteria for orchestrated trading agents; our Pydantic stage outputs and wide schedule CSVs are designed to support that traceability (outlook → stance → `final_signal`), even though we do not formally score policy convergence.

Benchmarks, simulators, and live evaluation. Qian et al. (2025) introduce the **Agent Market Arena (AMA)**, a lifelong multi-market benchmark showing that agent architecture often matters more than LLM backbone choice, a motivation for holding the model fixed (`qwen/qwen3.5-flash-02-23`) and varying regime context in §4. Papadakis et al. (2025) (**StockSim**) define order-level simulation with latency, slippage, and limit-order-book microstructure; our paper ledger is candlestick-level with stated fees and makes no LOB claim, treating StockSim-class realism as aspirational future work.

Empirical scope vs frontier benchmarks. Jeon and Lee (2026), Li et al. (2025), and Miyazaki et al. (2026) optimize for statistical stress tests across index universes, seeds, or simulated meetings. Hindsight optimizes for auditability and temporal integrity on a two-ticker, four-window case study (Table 1), so the comparison there is qualitative.

Positioning summary. Where TradingAgents supplies the graph skeleton and BlindTrade / Nguyen and Pham (2026) articulate *what to measure*, Hindsight implements *how to reproduce it*: simulation end dates, structured artifacts, net returns, and frozen CSV schedules that regenerate Tables 1–6 without rerunning LLM inference.

3. Method / System

3.1 Architecture

TradingAgentsGraph: **Analysts → Bull/Bear Researchers debate → trader → risk → final decision**. Core routing follows TradingAgents (Xiao et al., 2024); our modifications focus on temporal data policy, schema-constrained outputs, and the backtest/evaluation pipeline.

3.2 Analyst task decomposition

Each analyst role is bound to specific tools and emits a structured AnalystReport (outlook + narrative). This supports traceable rationale from raw data to downstream debate not coarse role labels alone.

Stage	Agent	Tools	Concrete tasks
Market	Market analyst	`get_stock_data` `get_indicators`	Fetch OHLCV; from a fixed 13-indicator Tier-1 catalog (50/200 SMA, 10 EMA, MACD line/signal/histogram, RSI, Bollinger middle/upper/lower, ATR, VWMA, MFI), select up to 8 complementary series
Fundamentals	Fundamentals analyst	`get_fundamentals` `get_balance_sheet` `get_cashflow` `get_income_statement`	Pull statements and issuer profile; analyze revenue, margins, balance-sheet trends
News	News analyst	`get_news` `get_global_news`	Company-targeted and macro news over a rolling window
Sentiment	Social-media analyst	`get_news`	Sentiment and social discussion synthesis (company/social queries)
Debate	Bull / Bear researchers	none	Argue buy vs sell thesis from shared evidence
Execution	Trader, Risk manager	none	Propose and gate BUY/HOLD/SELL

Structured fields are persisted in schedule CSVs for audit (`market_outlook`, `fundamentals_outlook`, `bull_implied_stance`, `final_signal`, etc.), supporting *explainable stability* in the sense of Dewansh (2026): a coherent trace from evidence to executable action.

3.3 Tools and structured outputs

Analyst tool calls respect the simulation end date: vendor queries are clamped so that no data after the as-of trade date is returned. Each stage emits a Pydantic-validated object, giving typed JSON artifacts.

3.4 Ticker anonymization

LLMs trained on public text encode firm-specific narratives tied to ticker symbols. In a point-in-time backtest, that pretraining leakage can bias decisions.

Hindsight anonymizes the issuer for all LLM-facing prompts and tool outputs:

1. **Deterministic ticker aliases.** `TickerMapper.for_real_ticker()` maps each real symbol to a stable synthetic id (e.g. `STOCK_####`).
2. **Round-trip vendor access.** Before each API call, tool wrappers call `resolve_ticker_for_vendor()`; responses pass through `scrub_ticker_text()`.
3. **News scrubbing.** `scrub_news_text()` replaces multi-word proper nouns with `PROPER_NOUN` placeholders (heuristic, news-only).

3.5 Backtest metrics

The paper ledger records daily agent equity after fees. Primary metrics emphasize **total return**, **maximum drawdown (MDD)**, and **fee drag**; risk-adjusted ratios are secondary on short windows (67–111 trading days).

Metric	Definition
Total return (net)	$((V_T - V_0)/V_0)$ on agent equity after fees
Gross return	Net return plus fees as a fraction of V_0
Fee drag	Gross return minus net return
Alpha vs B&H	Agent net return minus buy-and-hold return on the same adjusted closes
Max drawdown (MDD)	Peak-to-trough on agent equity; B&H MDD on $(V_0 \times \text{close}/\text{close}_0)$
Excess Sharpe ratio	Mean daily excess return (over risk-free) / stdev, $\times\sqrt{252}$.[1]
Sortino / Calmar	Implemented; reported in Table 5 (secondary)

3.6 Observability

Each trading day's graph run is traced with Langfuse via its LangChain callback handler: one trace per (ticker, trade date), keyed by a session id, covering every stage's LLM calls, tool calls, token counts, and latency.

[1] Risk-free rate is an India short-rate proxy of 6.5% p.a., converted to a daily rate $r_f/252$ and applied post-hoc on the frozen equity series. Annualization uses $\sqrt{252}$.

4. Case Study Evaluation

4.1 Evaluation scope and inferential limits

Dimension	This study	Typical frontier benchmark
Universe	2 liquid NSE large-caps (RELIANCE, TCS)	Index constituents (e.g., S&P 500, TOPIX 100)
Regime windows	4 pre-specified (2 tickers × bear/bull)	Full sample or rolling panels
Stochasticity	N=1 on three windows; N=5 on TCS bull	20–50 independent trials
Regime labels	Exogenous from underlying B&H (±10%)	Often endogenous or undisclosed
Bias control	Simulation end date + Strict anonymization	Strict anonymization (BlindTrade)
Costs	Stated delivery fees (net returns)	Often gross returns

Section 4 shows how one stack behaves under two contrasting trends per ticker. It is not a test of expected alpha, and results from four windows should not be extrapolated to a broad universe (§5).

Pre-specification. Calendar windows are listed in `regime-timeranges` before agent backtests. Regime labels are assigned exogenously from underlying buy-and-hold return and are independent of internal Bull/Bear Researcher debate roles.

4.2 Regime definition and protocol

Regime label	Rule (underlying B&H on processed window, per ticker)
Bear	Total return $\leq -10\%$
Bull	Total return $\geq +10\%$

Windows differ by ticker (e.g., TCS bull Jun–Sep 2024; RELIANCE bull Jan–Jun 2025). Each run uses $100,000 initial cash.

Ticker	Regime	Processed window
RELIANCE.NS	Bear	2024-08-01 .. 2025-01-03
RELIANCE.NS	Bull	2025-01-01 .. 2025-06-03
TCS.NS	Bear	2024-12-02 .. 2025-04-03
TCS.NS	Bull	2024-06-03 .. 2024-09-03

4.3 Positioning vs prior agent-trading systems

Table 1. Qualitative comparison (see §4.1 for scope limits).

Feature	Hindsight	BlindTrade	QuantAgents	Expert Investment Teams
Primary goal	Reproducible PIT stack + audit trail	Anonymized RL/GNN policy	Simulated agent meetings	Hierarchical analyst teams
Architecture	LangGraph (staged)	GNN + RL	Meeting simulation	Manager–analyst hierarchy
Stochasticity reported	N=1; N=5 on TCS bull	20 seeds	Multi-year + live tracking	50 trials
Universe	2 NSE large-caps	S&P 500 (PIT)	Multi-asset simulation	TOPIX 100
Bias control	Sim. end date + strict anonymization	Strict anonymization	Simulation feedback	Knowledge cut-off controls
Costs in backtest	Net (Zerodha delivery)	Varies	Varies	Varies
Key §4 metrics	Return, MDD, fee drag	Sharpe (mean ± std)	ARR, volatility	Median Sharpe

4.4 Results

Table 2. Agent vs buy-and-hold (frozen CSVs, $100,000 start). Returns and MDD are recomputed from daily `equity` and `close`.

(a) Returns. Net, gross, fee drag, and alpha vs buy-and-hold.

Ticker	Regime	Days	B&H	Agent (net)	Gross	Fee drag	α vs B&H
RELIANCE	Bear	111	−17.43%	−7.54%	−6.53%	1.01 pp	+9.89 pp
RELIANCE	Bull	108	+15.09%	+5.56%	+6.19%	0.64 pp	−9.53 pp
TCS	Bear	88	−20.42%	−4.83%	−2.92%	1.91 pp	+15.59 pp
TCS	Bull	67	+21.86%	+19.61%	+21.80%	2.20 pp	−2.25 pp

(b) Drawdown, fees, and signals. MDD is peak-to-trough. Fees are in dollars. B/H/S is buy/hold/sell days.

Ticker	Regime	B&H MDD	Agent MDD	Fees ($)	B/H/S
RELIANCE	Bear	21.0%	5.11%	1,010	7/28/76
RELIANCE	Bull	11.0%	2.80%	635	9/25/74
TCS	Bear	23.9%	6.78%	1,910	12/24/52
TCS	Bull	5.5%	1.65%	2,198	13/14/40

A 10/50 moving-average rule is a price-only baseline on the same four windows: long when the 10-day average is above the 50-day, otherwise flat. It uses the same ledger ($100,000, Zerodha delivery fees, no slippage) and no language model.

Table 3. 10/50 SMA versus the Table 2 agent (net return and alpha vs buy-and-hold).

Ticker	Regime	SMA net	SMA α	Agent net	Agent α
RELIANCE	Bear	−1.29%	+16.14 pp	−7.54%	+9.89 pp
RELIANCE	Bull	−4.43%	−19.51 pp	+5.56%	−9.53 pp
TCS	Bear	−4.07%	+16.35 pp	−4.83%	+15.59 pp
TCS	Bull	+13.31%	−8.55 pp	+19.61%	−2.25 pp

In both bear windows the rule also loses less than buy-and-hold, and by more than the agent. In both bull windows it trails buy-and-hold. The frozen TCS agent beats the rule; the five repeat runs do not (mean +7.79%; Table 4).

The TCS bull row in Table 2 is one trajectory. Table 4 gives five more runs of that window.

Table 4. Five repeat runs of the TCS bull window ($100,000 start, 64 days). Buy-and-hold is +21.86% on every run. The Table 2 trajectory is not included. Mean and SD are across these five.

Run	Days	Net	α vs B&H	MDD	B/H/S
3	64	+8.23%	−13.63 pp	2.68%	8/43/13
4	64	+4.89%	−16.97 pp	3.11%	4/39/21
5	64	+5.15%	−16.72 pp	1.81%	11/41/12
6	64	+8.64%	−13.22 pp	1.42%	11/33/20
7	64	+12.02%	−9.84 pp	2.74%	8/41/15
Mean		+7.79%	−14.08 pp	2.35%	
SD		2.92 pp	2.93 pp	0.71 pp	

Table 5. Secondary risk-adjusted metrics (post-hoc on frozen daily equity; excess Sharpe uses 6.5% p.a. risk-free rate).

Excess Sharpe is the agent or buy-and-hold equity path. Sortino and Calmar are on the agent path only.

Ticker	Regime	Excess Sharpe (agent)	Excess Sharpe (B&H)	Sortino	Calmar
RELIANCE	Bear	−4.27	−2.51	−3.10	−0.96
RELIANCE	Bull	+0.74	+1.25	+3.67	+1.94
TCS	Bear	−2.69	−3.13	−2.25	−0.71
TCS	Bull	+3.86	+3.27	+24.22	+11.88

Short windows and discrete daily rebalancing make risk ratios noisy; we interpret Table 5 alongside Table 2, not in isolation. RELIANCE bull is positive on total

return (+5.6%) with positive excess Sharpe (+0.74) but lags B&H on alpha (−9.53 pp).

Cross-ticker patterns (observed in these windows).

- **Bear (2/2):** agent loses less than B&H on total return and shows markedly lower MDD (e.g., RELIANCE bear: 5.1% vs 21.0%). Both runs are SELL-heavy, consistent with reduced long exposure during drawdowns.
- **Bull (mixed):** RELIANCE lags (+5.6% vs +15.1%, α −9.53 pp) with 74 SELL vs 9 BUY days. On TCS the frozen run nearly tracks buy-and-hold (−2.3 pp); the five repeat runs do not (mean −14.1 pp; Table 4).

Table 6. Selected outlook / stance shares (% of trading days). Full pivot in notebook.

Market columns are the market analyst's outlook. Stance columns are the bull researcher's buy share and the bear researcher's sell share.

Ticker	Regime	Market bullish	Market bearish	Bull stance buy	Bear stance sell
RELIANCE	Bear	10.8	31.5	100.0	88.3
RELIANCE	Bull	24.1	19.4	94.4	77.8
TCS	Bear	22.7	33.0	71.6	53.4
TCS	Bull	46.3	14.9	64.2	44.8

Internal bull researcher stances can remain buy-leaning while final signals stay defensive; downstream trader and risk stages drive executable actions.

4.5 Transaction-cost breakeven and architectural boundaries

Fee drag vs alpha. Fee drag is modest (0.6–2.2 pp of initial capital; Table 2). It does not explain RELIANCE bull under-participation: the −9.53 pp alpha gap dominates 0.64 pp fee drag. In bear windows, defensive positioning yields positive alpha (+9.89 pp REL, +15.59 pp TCS) despite similar fee levels, suggesting exposure management, not costs, drives regime-split outcomes. A 10/50 SMA rule shows the same bear-window pattern (Table 3), so the smaller bear losses are not specific to the agent.

Simplified breakeven spread. Let K be non-HOLD signal days and F total fees. A coarse **fee-and-turnover breakeven** is the average daily edge required to justify rebalancing: $F/(V_0 \cdot K)$. For RELIANCE bull ($K = 83$ actionable days, $F = \$635$), this is ≈7.7 bps/day, far below the 88 bps/day implied alpha shortfall (9.53%/108 days). The binding constraint is SELL-heavy coordination (74 SELL days), not delivery fees alone.

Architectural boundary (hypothesis). Multi-stage debate → trader → risk can compress drawdown in bear windows but cap rally participation in bull windows when final policy stays defensive despite buy-leaning researcher stances

(Table 6). RELIANCE bull lags by 9.5 pp on its single run. TCS bull lags on average: mean alpha −14.1 pp across five repeat runs, against −2.3 pp for the frozen run alone (Table 4). We read this as a regime-by-issuer interaction within one architecture.

We do not estimate latency-driven slippage (full Coordination Breakeven Spread per Nguyen and Pham, 2026); that remains future work alongside StockSim-class order-level simulation (Papadakis et al., 2025).

4.6 Figures

`scripts/generate_paper_figures.py` regenerates Figures 1–4 from the frozen schedule CSVs. Each TCS bull panel is that frozen trajectory (Table 2), not the five-run mean (Table 4).

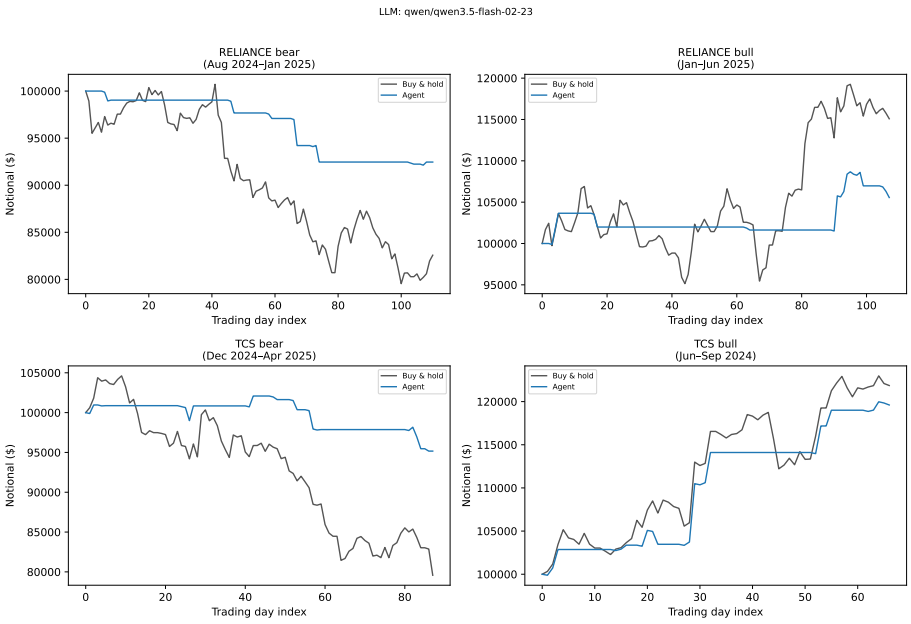


Figure 1: Agent vs buy-and-hold equity, one panel per ticker–regime window ($100,000 start).

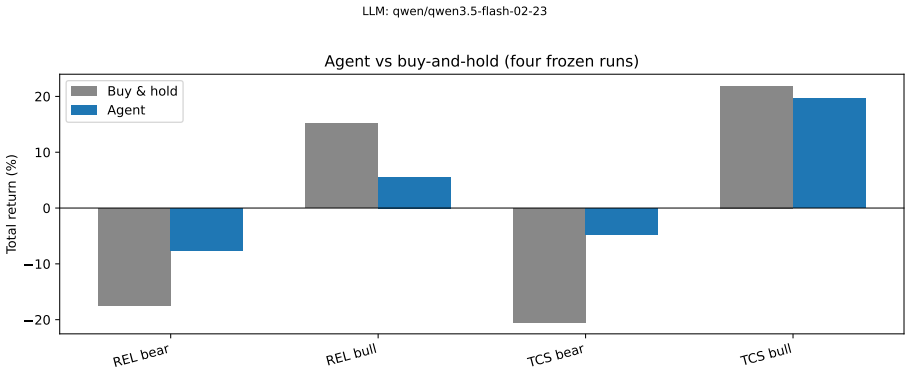


Figure 2: Total return, agent (net) vs buy-and-hold, four windows.

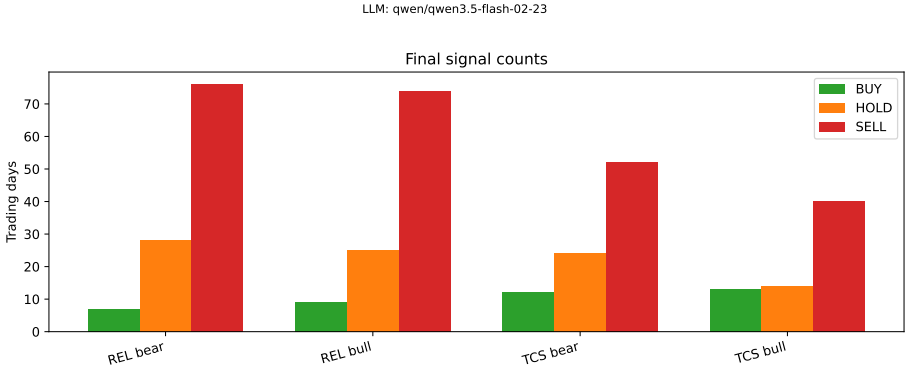


Figure 3: BUY/HOLD/SELL day counts per window.

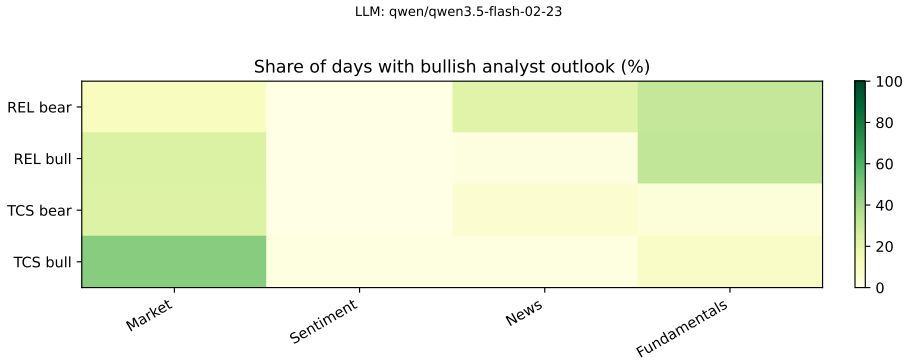


Figure 4: Share of trading days with a bullish analyst outlook, per window.

5. Conclusion

We built **Hindsight**, a point-in-time LangGraph stack with structured outputs, ticker anonymization on every run, and fee-aware backtesting, and ran it on RELIANCE.NS and TCS.NS. In the two bear windows the agent loses less than buy-and-hold and draws down less. A 10/50 moving-average rule does the same. In the two bull windows the agent trails buy-and-hold: by 9.5 pp on the single RELIANCE run, and on TCS by 2.3 pp in the first run and 14.1 pp on average across five further runs.

Limitations and future work.

- **Inferential scope:** N=1 on three windows; N=5 repeat runs on TCS bull only; two tickers, not an index universe.
- **Calendars:** Per-ticker windows are not synchronous; no shared macro calendar claim.
- **Model and ablations:** Single LLM (`qwen/qwen3.5-flash-02-23`); ticker anonymization on for every reported run; reduced-pipeline presets (market analyst only; all analysts, no debate; debate, no risk stage) exist but were not run for §4.
- **Metrics:** Short windows; risk ratios are secondary; no latency/slippage CBS.
- **Future empirical work (out of scope here):** multi-seed trials on the other three windows, broader PIT universes, and full CBS with measured agent latency.

Regimes are exogenous labels, not agent predictions. **Not claimed:** SOTA returns; regime forecasting; seed variance beyond TCS bull; ablation results; statistical generalization beyond the four windows.

Contributions

1. **Reproducible backtest harness and frozen artifacts**: fee-aware paper ledger, schedule CSVs, replication scripts.
2. **Point-in-time data policy and mandatory anonymization**: simulation end date at vendor routing; deterministic ticker aliases.
3. **Structured stage outputs**: Pydantic-constrained artifacts for audit and traceability.
4. **Multi-agent decision graph** with tool-bound analyst decomposition.
5. **Regime-split case study**: in bear windows the agent loses less than buy-and-hold and draws down less; in bull windows it trails. Five further runs on TCS bull, and a 10/50 moving-average baseline.

Acknowledgements

The agent graph implementation builds on TradingAgents (Xiao et al., 2024; Tauric Research, Apache License 2.0). We thank the authors for open-sourcing the framework.

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